A Wideband Low-Cost Reconfigurable Reflectarray Antenna With 1-Bit Resolution

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***Abstract*— A wideband 1 bit reflectarray antenna (RA) with two-dimensional (2-D) electronically beam-scanning capability is presented in this article. Based on the basic structure of the magnetoelectric (ME) dipole antenna, a novel reconfigurable RA (RRA) element is proposed for wideband operation by combining electric dipole resonance and magnetic dipole resonance. Each element is integrated with only one p-i-n diode to change the types of resonances, leading to a reconfigurable reflecting phase with a 1 bit resolution. As a demonstration, a wideband electronically RRA with 16 × 16 elements is developed, fabricated, and measured. In the proposed RRA, each element is connected with a separate bias line to independently control the working state and reflecting phase. Measured results show that the proposed RRA can achieve a 3 dB gain bandwidth of 38.4% with a peak aperture efficiency of 24% at 12 GHz. Besides, the proposed antenna can support the wide scanning angles ranging from −60° to +60° in the E-plane and 0° to +60° in the H-plane within the whole operating bandwidth. The proposed simple design could be a promising candidate for intelligent reflective surface (IRS) applications.**

***Index Terms*— Beam scanning, low cost, magnetoelectric (ME) dipole antenna, reconfigurable, reflectarray, wideband.**

I. Introduction

OVER the past ten years, there has been a great demand for high-gain antennas to compensate for the high propagation loss at higher operating frequencies. However, high gain of the antenna usually means narrow beamwidth and small spatial coverage, leading to the desire for the beam-scanning capability. Besides, to satisfy the increasing data speed, wide operating bandwidth is usually in demand to support more communication channels. Therefore, wideband high-gain antennas with beam-scanning ability would be promising candidates for next-generation wireless communication, remote sensing, and radar systems [1], [2].

The phased array is a classical method to achieve the high-gain property and beam-scanning capability [3]–[5]. By applying different excitation phases for each element, the radiation beam can be steered in two dimensions (2-D) with high performance. However, complicated feeding networks are required in these analog designs [3], [4], which increases the design complexity and power consumption. Although some digital phased arrays have been implemented, the high cost and bulky size impede their applications [5]. Beam scanning can also be achieved by mechanically rotating the reflectors in the parabolic reflector antennas [6]. This type of antenna has a simple structure with low design complexity, but the scanning speed is slow and a servo system is required.

Recently, the reflectarray antenna (RA) has received tremendous research interest since it combines the advantages of the reflector antenna and phased array antenna [7], [8]. Due to the spatial feeding mechanism, the complex and lossy feeding network can be eliminated for performance enhancement, structural simplicity, and fabrication robustness. Specifically, the reconfigurable RA (RRA) with dynamic beam-scanning ability has also been widely investigated and reported. Based on the mechanically reconfigurable mechanism, height-tunable patch [9] and rotational split rings [10] are proposed for LP and CP designs with the assistance of RF-MEMS technology. For beam-scanning applications in higher operating frequencies, tunable materials are employed, including the liquid crystal, GeTe material, and graphene [8]. Nevertheless, the electronically RRA is more popular due to its fast scanning capability when compared with its counterparts. In this method, electrical components, including p-i-n diodes [11], [21], RF-MEMS switches [22], and varactors [23], [24], are employed to obtain the discrete or continuous phase compensation schemes.

Due to their simplicity and low cost, electronically RRAs with the 1 bit resolution are drawing practical interest. Although the 3 dB phase quantization loss is observed in the antenna gain, the 1 bit RRAs can achieve satisfactory scanning performance, especially when a larger aperture size is applied [25], [26]. Several reported works have demonstrated the merits of the 1 bit RRAs. By integrating with four and two p-i-n diodes, frequency-selective surface (FSS) structures are used for polarization diversity [11] and bidirectional operation [12]. Meanwhile, patch antenna is also widely used as the unit cell in the RRAs due to its simple structure. Based on the receiving-and-transmitting method, 1 bit resolution can be achieved by soldering two p-i-n diodes on the feeding network

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to switch the direction of the E-fields [13]. Circular patch [14] and square patch [15] are introduced for circular polarization and folded applications by incorporating two orthogonally arranged p-i-n diodes.

However, these designs have to employ more than one p-i-n diode within a single element. If the aperture size becomes large, the complexity of the bias circuit and the cost will dramatically increase. To reduce the number of p-i-n diodes used in each element, the radiating edge of the rectangular patch is soldered with a single p-i-n diode [16]–[19]. By applying different voltages, the p-i-n diode can be switched ON or OFF. Therefore, the working mode of the patch can be switched to realize a 180° reflecting phase difference. The p-i-n diode is also placed at the backside of the ground plane [20] to reduce its scattering effect. For bandwidth enhancement, two separated patches are connected by a p-i-n diode to evoke a 180° phase shift [21]. Unfortunately, the operating bandwidth of all the reported electronically reconfigurable designs is still narrower than 20% due to the intrinsic characteristic of the elements. Therefore, it is still a challenging work to design a low-cost RRA with wide operating bandwidth.

In this article, the concept of ME dipole antenna is introduced into the linearly polarized RRA element. The proposed RRA element can achieve wide operating bandwidth by combining the electric dipole resonance and magnetic dipole resonance. For the first time, the types of both resonances are changed accordingly by a single p-i-n diode to achieve a 180° reflection phase difference (1 bit resolution). As a demonstration, an electronic RRA with 16 × 16 elements is proposed to obtain the wide-angle beam-scanning capability. Moreover, the proposed RRA design can achieve a much wider operating bandwidth than the reported electronically RRAs.

The rest of this article is organized as follows. In Section II, the design and operating principle of the wideband 1 bit RRA element are demonstrated. Then, the phase compensation strategy and the performance of the proposed RRA are discussed in Section III. Finally, a conclusion is drawn in Section IV.

II. Design of the Wideband RRA Element

A. Structure of the 1 bit RRA Element

The ME dipole antenna has attracted much research interest since it was proposed [27] due to its simple structure and good radiation performance, such as wideband operating bandwidth, stable high gain, and symmetrical radiation patterns. The basic structure of substrate integrated ME dipole is shown in Fig. 1 [28]–[30]. Two pairs of patches are placed horizontally to work as the electric (E) dipole, while the adjacent shorting vias and the ground plane work together as a shorted patch and radiate as a magnetic (M) dipole, which requires the height of the substrate to be about one quarter wavelength [28]. An L-shaped feeding probe is used to excite the E dipole and M dipole simultaneously. Therefore, the wide operating bandwidth can be easily achieved by combining the two resonances.

Based on the basic ME dipole structure, a wideband RRA element is proposed and its configuration is shown in Fig. 2. It mainly consists of two parts: the ME dipole part and the bias

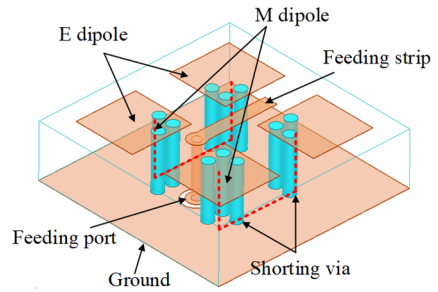


Fig. 1. Configuration of the conventional L-shaped probe-fed ME dipole antenna.

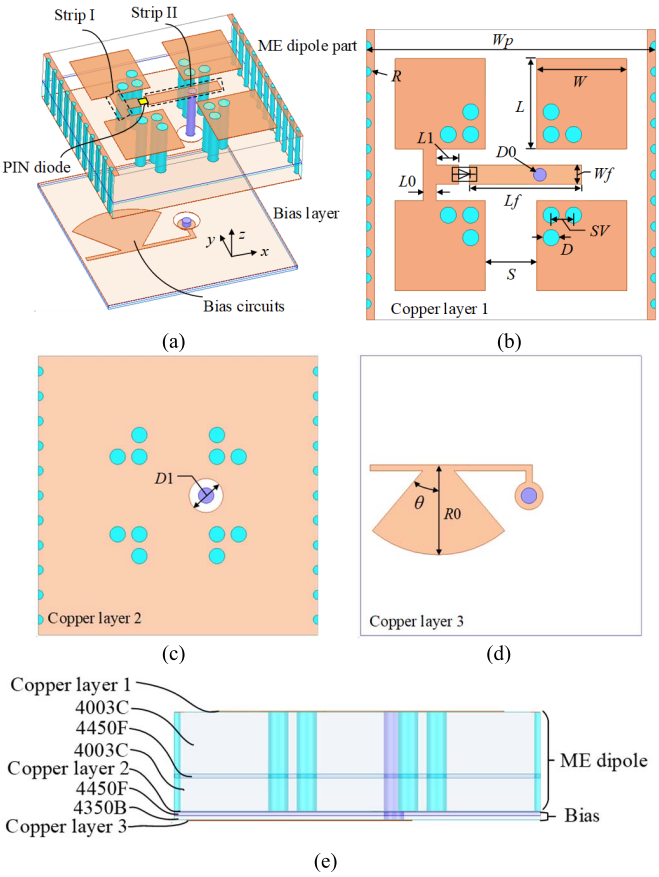


Fig. 2. (a) Configuration of the proposed wideband 1 bit RRA element. (b)–(d) Copper patterns on each cooper layer. (e) Stack up of the proposed RRA element.

part. The structure of the ME dipole part is similar to that of the conventional ME dipole antenna. For the desired operating band (10–15 GHz), the optimized height of the substrate is 2.42 mm using Rogers RO4003C, which is about $0.22\lambda_g$ (λ_g is the wavelength in dielectric at 14 GHz). To achieve this desired profile, two substrate layers (Rogers RO4003C, dielectric constant of 3.55, and thickness of 0.813 and 1.524 mm) are bonded together by a bonding film (Rogers RO4450F, dielectric constant of 3.52, and thickness of 0.101 mm). Two columns of shorting vias are introduced perpendicularly to the exciting electric fields.

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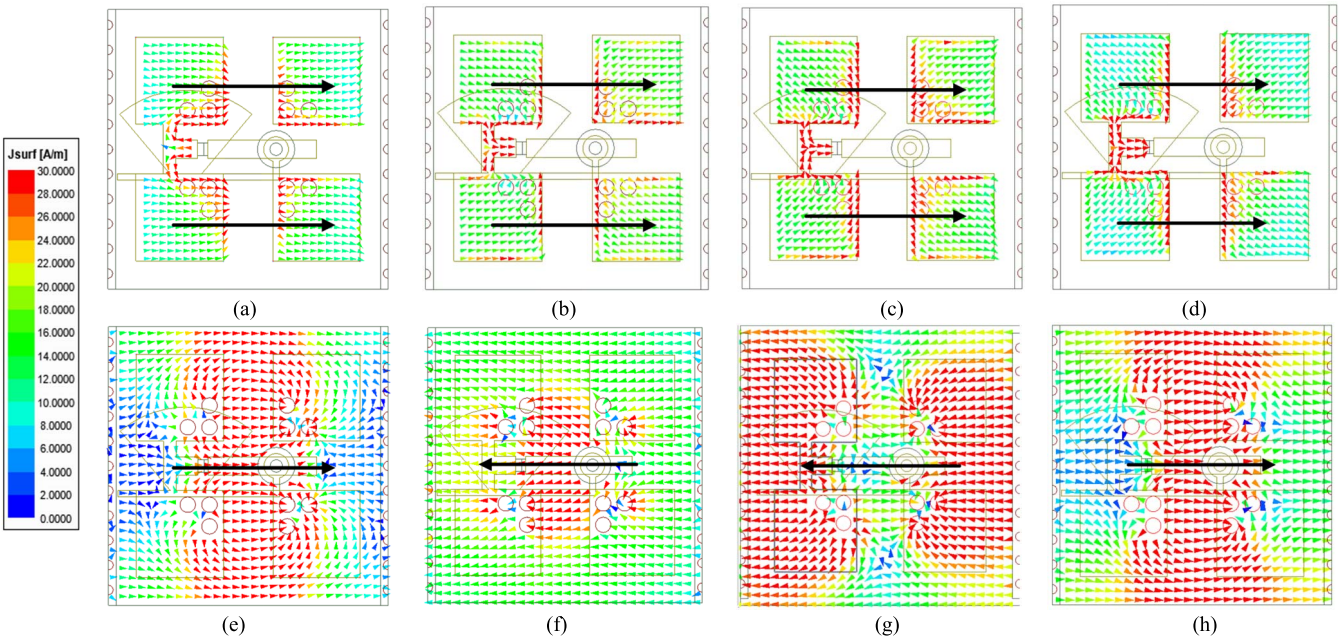


Fig. 3. Current distributions on the patches and the ground planes. (a) Current on the patch at 10 GHz for State I. (b) Current on the patch at 14 GHz for State I. (c) Current on the patch at 10 GHz for State II. (d) Current on the patch at 14 GHz for State II. (e) Current on the ground at 10 GHz for State I. (f) Current on the ground at 14 GHz for State I. (g) Current on the ground at 10 GHz for State II. (h) Current on the ground at 14 GHz for State II.

To realize reconfigurability, two narrow strips are introduced for the arrangement of p-i-n diode, denoted as Strips I and II in Fig. 2(a). The two patches on the left side are connected with each other by Strip I. Since the unit cell is illuminated by the x-polarized electromagnetic wave, only the currents along the x-axis will be excited. Therefore, the introduction of Strip I has negligible effects on the RF performance. However, Strip I can be treated as a shorting point for dc current since the patches and the ground are connected with each other by the four pairs of shorting vias. Strip II with a width of *Wf* and a length of *Lf* is employed along the x-axis. The cathode and anode of a p-i-n diode are soldered on Strips I and II, as shown in Fig. 2(b). As mentioned before, Strip I is connected with the ground plane. Therefore, its dc voltage is always 0 V. To provide the bias voltage for Strip II, an additional metallic via with a diameter of D_0 connects Strip II with the bias circuit. A clear hole is etched on the ground plane to isolate the ground and the dc bias metallic via. The bias part is based on a single substrate (Rogers RO4350B, dielectric constant of 3.55, and thickness of 0.101 mm) and the bias circuits are etched on the backside of the substrate. A fan-shaped open stub is used in the bias circuit to work as the low-pass filter. This structure can isolate the RF signal and the dc signal over a wide bandwidth. Therefore, the amplitude of the E-fields at the biasing point is always nearly 0. As a result, the influence of the dc circuit can be ignored. Finally, the ME dipole layer and the bias layer are bonded by another bonding films.

B. Working Principle of the 1 bit RRA Element

When different voltages are employed on the dc bias line, the p-i-n diode can be switched ON or OFF. As a result, the left side of Strip II will connect with the shorting point if the p-i-n diode is switched ON, while it will remain an open point if the p-i-n diode is switched OFF. Therefore, different reflecting phase responses can be obtained for these two states. To reveal the operating principle of the proposed wideband RRA element, numerical studies are carried out using the full-wave solver ANSYS HFSS. In the simulation, periodic boundary and Floquet ports are assigned to the RRA element.

Fig. 3 shows the current distribution on the patches and ground plane when the element is illuminated by normal x-polarized waves. State I denotes the circumstance when Strip II is biased with positive voltages and the p-i-n diode is switched OFF. State II is the circumstance when Strip II is biased with a negative voltage and the p-i-n diode is switched ON. As can be observed from Fig. 3, when the element works under State I, the currents on the patches and ground plane are in the same direction at 10 GHz. Therefore, the whole structure works as an electric dipole resonance and acts as an electric wall with a 180° reflecting phase, while at 14 GHz, the currents on the patches and ground plane are in opposite directions. This means that the element performs like a magnetic dipole resonance and act as a magnetic wall with a 0° reflecting phase [31]–[32]. On the contrary, when the p-i-n diode is switched ON and the element works under State II, the currents on the patches and ground are in opposite directions at 10 GHz, introducing a magnetic dipole resonance and a 0° reflection phase. Meanwhile, the currents are in the same direction at 14 GHz, resulting in an electric dipole resonance with a 180° reflection phase. The working resonant modes and the corresponding reflecting phases of the proposed RRA element are summarized in Table I.

According to the analysis above, the proposed RRA can support an electric dipole resonance and a magnetic dipole resonance simultaneously to broaden the working bandwidth.

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TABLE I

TYPE OF RESONANCE AND REFLECTING PHASE RESPONSE OF THE PROPOSED WIDEBAND RRA ELEMENT UNDER THE DIFFERENT WORKING STATES

Working State	10GHz		14GHz	
	Type of resonance	Reflecting phase	Type of resonanc	Reflecting phase
State I (OFF)	electric	180°	magnetic	0°
State II (ON)	magnetic	0°	electric	180°

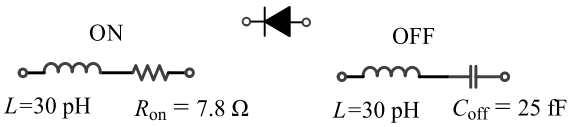


Fig. 4. Equivalent circuits of the p-i-n diodes under different working states.

TABLE II

DIMENSIONS OF THE WIDEBAND 1 BIT RECONFIGUABLE REFLECTARRAY ANTENNA ELEMENT (UNITS: mm)

Parameters	*W*	*L*	*Wf*	*Lf*	*L*0	*L*1
Values	2.8	2.8	0.6	3.5	0.4	0.7
Parameters	***S***	***SV***	***D***	***D*0**	***D*1**	***R***
Values	1.6	0.7	0.5	0.4	1.2	0.15
Parameters	***R*0**	***θ***	***Wp***			
Values	3.3	40deg	9			

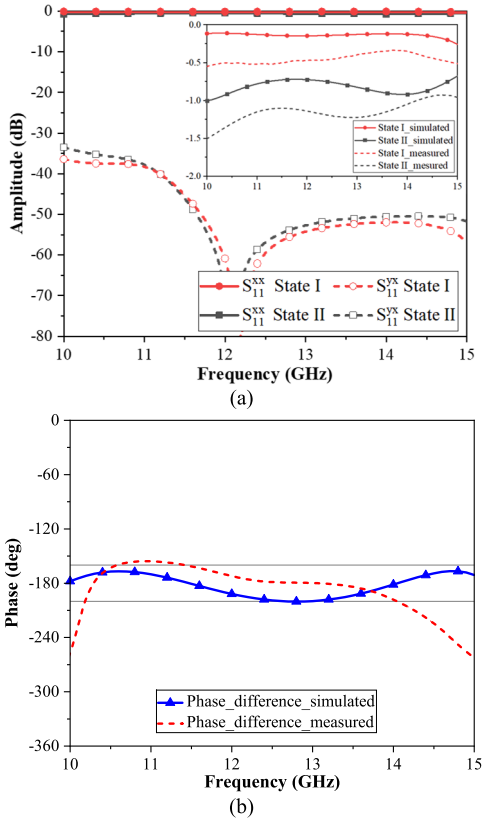


Fig. 5. Simulated and measured performances of the proposed wideband RRA element under the normal x-polarized incident wave with different bias voltages. (a) Reflection amplitudes of S_{11}^{xx} and S_{11}^{yx}. (b) Reflection phase response of S_{11}^{xx}.

Besides, the types of both two resonances can be changed easily by switching the working state of the p-i-n diode, resulting in a 180° phase difference within a wide operating bandwidth. It is also observed from Fig. 3 that the current on the patches and the ground are almost along the x-direction if the element is illuminated by the x-polarized waves. This property implies that the proposed RRA element can realize high polarization isolation.

C. Performance of the 1 bit RRA Element

In this article, MACOM MADP-000907 is selected as the p-i-n diode to control the reflecting phase. The equivalent circuits of the p-i-n diode are shown in Fig. 4. At the operating bandwidth, the p-i-n diode can be modeled as a series connection of a resistance of $R_{\mathrm{ON}} = 7.8\ \Omega$ and an inductance of $L = 30$ pH for the ON-state, and a series connection of a capacitance $C_{\mathrm{OFF}} = 25$ fF and an inductance of $L = 30$ pH for the OFF-state [17].

Table II shows the parameters of the proposed 1 bit RRA element. The periodicity of the unit cell is $Wp = 9$ mm, which is $0.375\lambda_0$ (λ_0 is the wavelength in free space at the center frequency $f = 12.5$ GHz) to avoid the grating lobes at the high operating frequency.

Fig. 5(a) shows the simulated and measured results of the proposed 1 bit RRA element under the normal x-polarized incident wave. The measurement method is demonstrated in [33]. The measured insertion losses of the two states are larger than the simulated ones. However, the measured insertion losses of the two states are both lower than 1.5 dB within the bandwidth of 40%, from 10 to 15 GHz. Also, the largest measured insertion losses are 0.55 and 1.5 dB for States I and II, respectively, which is acceptable for an active RRA element. The high insertion loss for State II is mainly introduced by the parasitic resistor of the p-i-n diode. Low cross-polarization levels can be observed within the whole operating bandwidth for different working states, indicating the high polarization isolation of the proposed RRA element.

The simulated and measured reflecting phase differences of the two states are shown in Fig. 5(b). Within the working bandwidth, the measured reflecting phase difference of the two states agrees well with the simulated one, fluctuating along 180° with a ripple of ±20°. The phase ripple will introduce an acceptable gain loss (about 0.2 dB) in the final reflectarray design [17].

To illustrate the functionality of shorting vias, Fig. 6 shows the simulated results of the RRA element with/without shorting vias. It can be observed that a dropped point is located in the working bandwidth due to the resonance of the RRA element [34], [35]. The inband resonance will introduce uncertainty in the performance of the final RRA. To address this problem, two columns of shorting vias are applied here to introduce an additional EM filed boundary condition and

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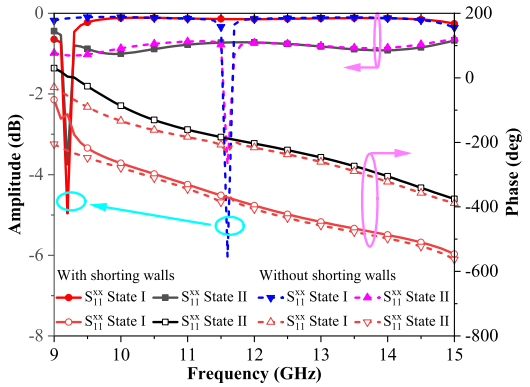


Fig. 6. Simulated performance of the proposed wideband RRA element with/without shorting vias.

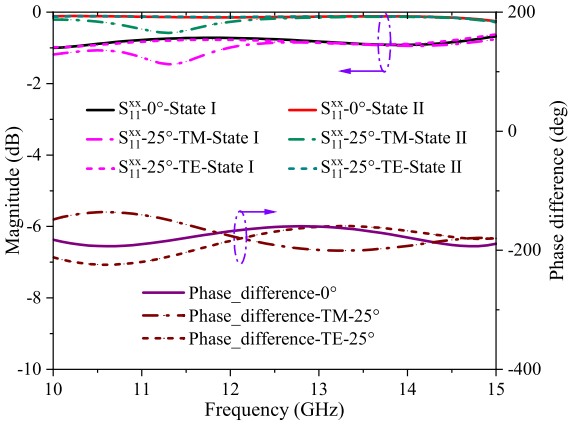


Fig. 7. Simulated performance of the proposed wideband RRA element under oblique x-polarized incident waves with different bias voltages.

reduce the mutual coupling of the adjacent element. It can be observed that after introducing the two columns of shorting vias, the resonance is shifted to 9.2G, leaving a smooth operating band from 10 to 15 GHz.

Fig. 7 shows the reflection coefficients under the oblique incident waves. When the element is illuminated by the TE waves, it can achieve a similar amplitude performance with that under the normal incident wave. However, for the TM wave, the insertion loss at the lower frequency is relatively higher but still lower than −1.5 dB for States I and II. The phase difference between the two states is also shown in Fig. 7. For TM waves, the phase difference fluctuates along 180° with a ripple of 20° within the operating bandwidth, but for TE wave, the phase ripple is a bit larger than 20°, which is still acceptable since only few edged elements are under this oblique incidence. Therefore, the element can achieve a relatively stable performance even under the oblique TE and TM waves. Thus, the reflectarray can be simply designed according to the performance under normal incidence.

III. Design of the Wideband RRA

A. Calculation of Phase Compensation Scheme

Basically, the RRA works as a spatial phase shifter to convert the spherical wave from the feed to a planar wave for

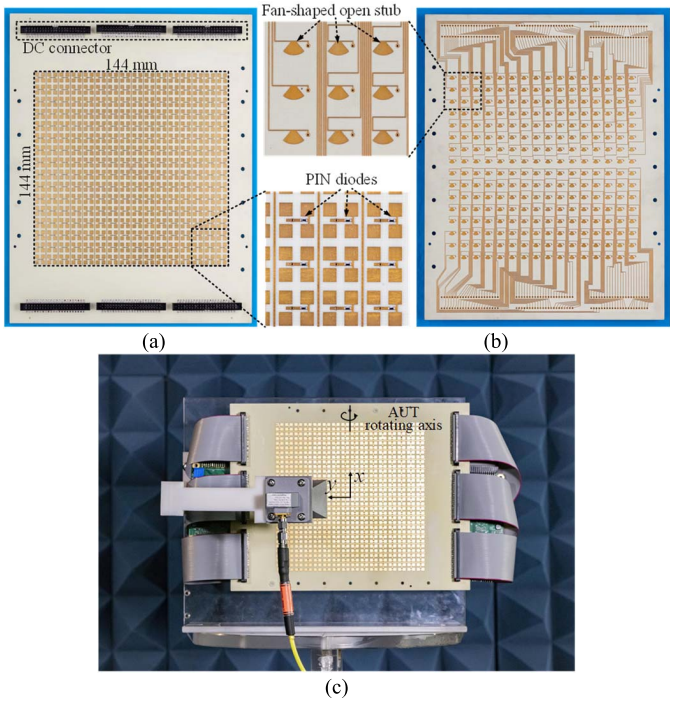


Fig. 8. Photograph of the proposed wideband reconfigurable reflectarray. (a) Top view. (b) Bottom view. (c) Prototype of the RRA under test of H-plane.

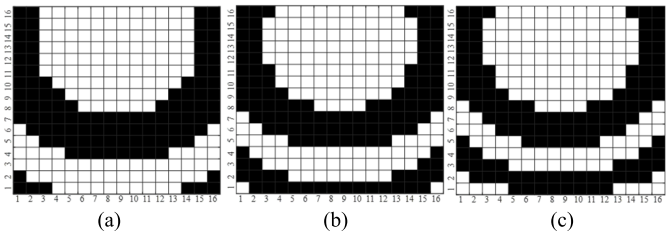


Fig. 9. Phase compensation schemes for broadside radiation patterns. (a) 11 GHz. (b) 12.5 GHz. (c) 15 GHz.

different scanning angles. Therefore, the desired compensation phase of each element should be deliberately designed.

To mitigate the blockage, a pyramidal horn with a 14 dBi gain at 12.5 GHz is offset by −15° to produce the x-polarized incident spherical waves. Besides, the main beam of the feeding horn is pointed to the center of the RRA aperture. The ratio of focal length to aperture diameter (f/D) is selected as 1.2 to guarantee the elements' performance under the oblique incident wave.

After deciding the position of the feeding horn and the size of the aperture, the initial desired compensation phase is calculated by

$$\varphi_{mn} = k_0\left(R_{mn} - \vec{r}_{mn}\cdot\vec{b}\right) + \varphi_0 \tag{1}$$

where k_0 is the propagation constant in free space, R_{mn} is the distance between the phase center of the feeding horn and the mnth element on the aperture. $\vec{r}_{\mathrm{mn}}$ is the position vector of the mnth element, and $\vec{b}$ is the unit vector of the desired output beam direction. φ_0 is the reference phase, which is

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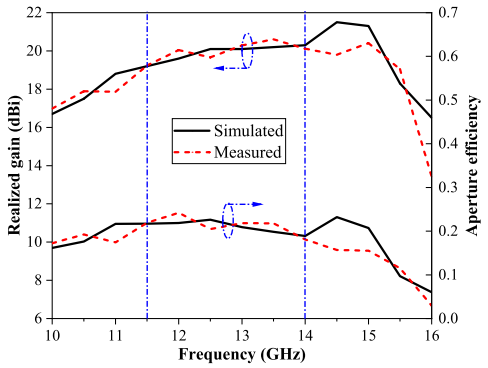


Fig. 10. Measured and simulated broadside gains of the proposed RRA under the different phase configurations in Fig. 9.

employed to recover the performance degradation introduced by the phase quantization scheme.

The strategy for assigning the phase-correction value for each element is expressed as follows:

$$\begin{cases} \varphi \le 90^\circ or \varphi > 270^\circ \rightarrow \varphi = 0^\circ \\ 90^\circ < \varphi \le 270^\circ \rightarrow \varphi = 180^\circ. \end{cases} \tag{2}$$

Since the proposed element can provide only two phase compensation states, there are always phase errors between the desired phase compensation and the quantized one, which deteriorates the performance. Therefore, the reference phase φ_0 is optimized to achieve the minimal phase error of the whole RRA aperture. The cost function used in this article is shown in the following equation:

$$CF = \sum_{i=1}^{M} \sum_{j=1}^{N} \cos^q \theta_e(i, j) \cdot \varphi_{error}(i, j) \tag{3}$$

where M and N represent the number of rows and columns in the array, respectively. $\varphi_{error}(i, j)$ is the absolute phase error at each reflectarray element. $\cos^q \theta_e(i,j)$ is the amplitude of the incident field for each element, where the radiation pattern of the feeding horn is modeled by the function $\cos^{2q}(\theta)$. Therefore, the compensation phase distribution for different scanning angles at different operating frequencies can be obtained by using (1)–(3).

B. Prototype and Experimental Result

Based on the proposed RRA element and the phase compensation strategy, a wideband 1 bit 16 × 16 RRA with 2-D dynamic beam-scanning capability is designed, fabricated, and measured. The prototype of the RRA is shown in Fig. 8. The RRA is realized by the printed circuit board (PCB) technology and the feed horn is fabricated by purely metallic 3-D printing. The p-i-n diodes are mounted on the top surface of the RRA by surface mounting technology (SMT). For testing, an FPGA-based logic digital circuit board is employed to control the p-i-n diode on each element individually through the bias lines. In each test round, the calculated quantized phase compensation scheme is loaded into the FPGA and translated to a voltage map with positive (+1.3 V) and negative voltages (−1.3 V) to control the reflecting phase of each RRA element.

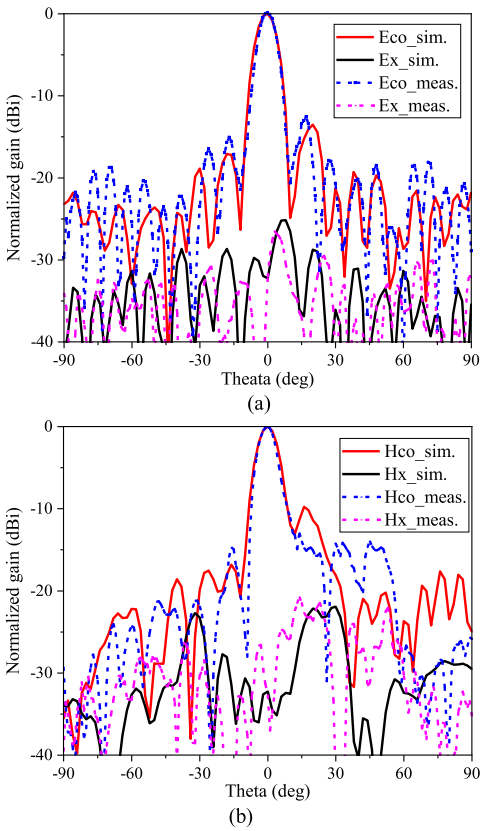


Fig. 11. Measured and simulated broadside radiation patterns at 12.5 GHz. (a) E-plane. (b) H-plane.

A coaxial cable to waveguide adapter (HD-120WCASKPC) was used to connect the feeding horn for measurement. The radiation performance of the proposed RRA was measured by utilizing a compact range far-field measurement system.

For the reported RRA with narrow bandwidth, the phase compensation scheme is calculated at the center frequency and different phase compensation schemes are assigned only for beam-scanning ability. This is acceptable since the phase errors at edge frequencies are small. However, for a wideband reflectarray, each frequency point has its own unique best phase compensation scheme. Therefore, even if the desired scanning angle is the same, different phase compensation schemes should be assigned for different operating frequency points to fully extend the potentials of wideband RRAs. Therefore, in measurement, different phase schemes are assigned to the RRA for both different frequency points and different beam-scanning angles. For simplicity, the broadside gain of the proposed RRA is measured under three frequency bands, i.e., 10−11.5, 11.5−14, and 14−16 GHz. The associated phase compensation schemes of three bands are shown in Fig. 9(a)–(c), which are calculated based on the frequencies of 11, 12.5, and 15 GHz, respectively. It should be mentioned that even with the same compensation schemes calculated based on the center frequency (f = 12.5 GHz), the RRA can also obtain a stable performance within a wide bandwidth.

Fig. 10 shows the measured and simulated broadside gains and the aperture efficiencies of the RRA under these

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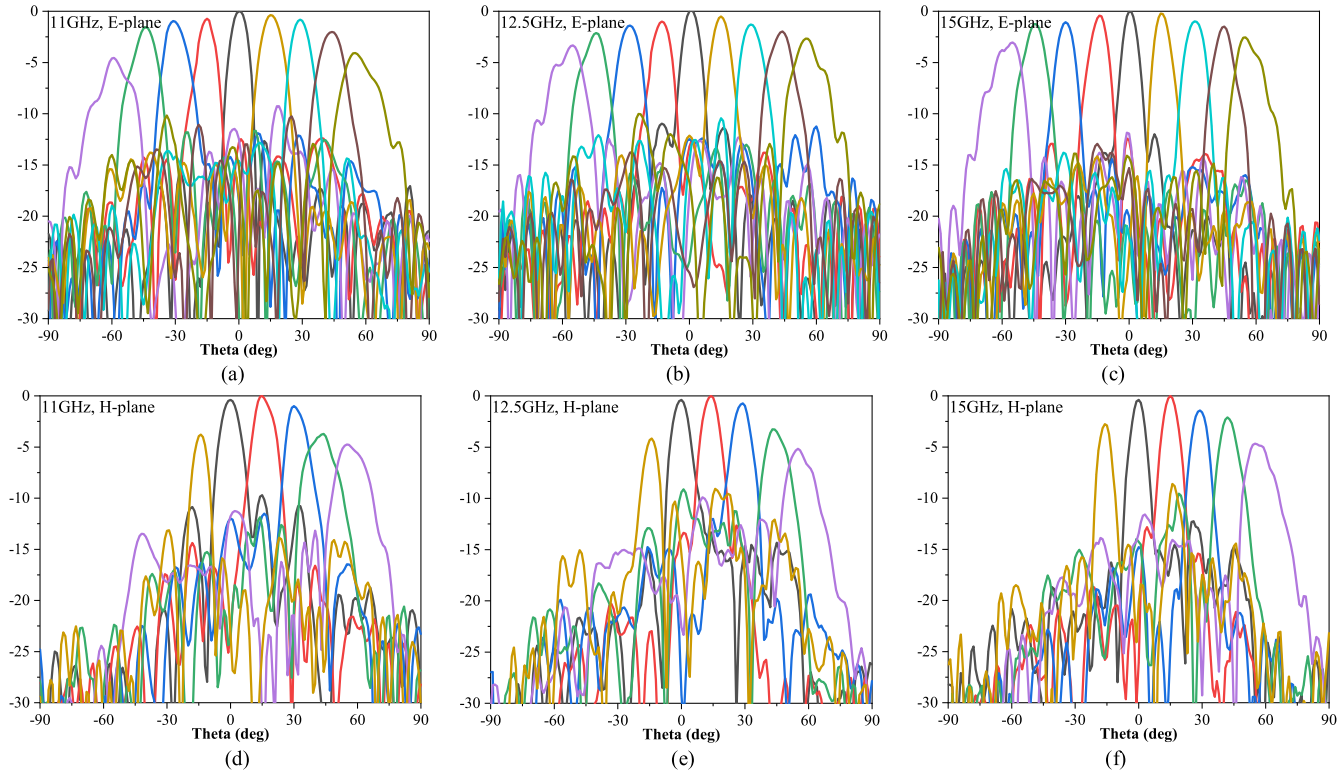


Fig. 12. Measured radiation patterns of the proposed array: (a) E-plane at 11 GHz, (b) E-plane at 12.5 GHz, and (c) E-plane at 15 GHz; and (d) H-plane at 11 GHz, (e) H-plane at 12.5 GHz, and (f) H-plane at 15 GHz.

TABLE III
MEASURED GAINS FOR DIFFERENT SCAN ANGLES

Scan angle			0°	15°	30°	45°	60°
Gain (dBi)	11	E-plane	17.8	17.1	16.9	16.3	13.3
		H-plane		18.2	17.1	14.5	12.3
	12.5	E-plane	19.7	18.9	18.4	17.6	16.3
		H-plane		20.1	19.3	17.1	14.4
	15	E-plane	20.4	19.9	19.4	19.1	17.4
		H-plane		21	19.6	18.7	15.6

three-phase compensation schemes in Fig. 9. Good agreements can be observed between the measured and simulated results. The measured gain ranges from 17.9 to 20.6 dBi within the operating bandwidth from 10.5 to 15.5 GHz, which agrees well with the simulated one. The measured 3-dB gain bandwidth is 38.4% and the peak aperture efficiency is 24%, corresponding antenna gain of 20.1 dBi at 12 GHz. The relatively low aperture efficiency is caused by the 1 bit phase compensation scheme and the ohmic loss of the p-i-n diode.

Fig. 11 shows the simulated and measured radiation patterns at 12.5 GHz with the phase compensation scheme for broadside radiation patterns. In the E-plane, the measured results agree well with the simulated ones. The main lobes and the locations of the sidelobes are well predicted. In contrast, some discrepancy between the simulated and measured results can be observed in the H-plane. This is mainly because the dc connectors and wires are placed at the H-plane and not included in the simulation. Therefore, their scattering effects deform the radiation patterns. This problem can be alleviated if these components are placed at the back of the RRA [14], [20]. The cross-polarization levels in the main beams are about −33 and −31.5 dB for simulation and measurement, respectively. This is benefitted from the low S_{11}^{yx} of the RRA element. Besides, the measured sidelobe levels (SLLs) in the E-plane and H-plane are −12.5 and −13.8 dB, respectively.

Fig. 12(a)–(c) shows the measured radiation patterns scanning from −60° to 60° with a progressive increment of 15° in the E-plane. It can be observed that the proposed RRA can obtain large scanning angles within the whole operating bandwidth, from 11 to 15 GHz. The pencil beams are obtained in the desired directions with low scanning losses and no grating lobes. The measured maximal scan losses in the E-plane are 4.5, 3.4, and 3 dBi for 11, 12.5, and 15 GHz, respectively, while in the backward directions of H-plane, the radiation patterns are deteriorated badly due to the blockage of the feeding horn. When the output beam is designed for $\theta = -15°$, the pencil beam is severely deformed, resulting in large scan loss and high SLLs. In the forward directions of the H-plane, the proposed RRA can also scan to 60° with scan losses of 5.5, 5.3, and 4.8 dBi at these three operating frequency points. Apart from the scan loss, the proposed RRA achieves better performance at higher operating frequency points in terms of the SLLs. This is because the feeding horn has lower directivity in the lower operating bandwidth, leading to a more uniform illumination on the RRA aperture. Besides, the phase quantization errors also contribute to the high SLLs [25], [26].

The detailed scanning performance of the proposed RRA is summarized in Table III. It should be mentioned that when the beam is scanned to $\theta = 15°$ in the H-plane, the obtained

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TABLE IV
COMPARISON BETWEEN THE PROPOSED AND REPORTED WIDEBAND 1 BIT RECONFIGURABLE REFLECTARRAYS

Ref.	Frequency (GHz)	Number of elements	Type of unit cell	Number of PIN Diode	Polarization	Peak Aperture efficiency (%)	Peak Gain (dBi)	Gain bandwidth (%)
[11]	13.5	10*10	M-FSS	4	LP/CP	<15	17.6	10 (1-dB)
[13]	12.5	16*16	Patch	2	LP	15.4	20.5	15.4 (1-dB)
[14]	9.5	16*16	Patch	2	CP	20	21.8	9.4 (3-dB)
[15]	12.5	16*16	Patch	2	LP	13.2	17.07	17 (3-dB)
[17]	12.5	10*10	Patch	1	LP	17.9	17.5	12 (3-dB)
[18]	60	160*160	Patch	1	LP	9.5	42	
[21]	5	12*12	Patch	1	LP	15.2	19.22	8.4 (1-dB)
This work	12.5	16*16	ME dipole	1	LP	24	20.1	38.4 (3-dB) 22.3 (1-dB)

antenna gain is relatively higher than that of $\theta = 0°$. This is mainly due to the specular reflection of the feeding horn. However, the specular reflection also causes high SLLs in several radiation patterns of H-plane.

C. Comparison and Discussion

The proposed RRA is compared with other reported 1 bit RRA in Table IV. The proposed RRA can achieve a much wider 3 dB gain bandwidth than other reported 1 bit electronically RRA. This is mainly beneficial from the wide operating bandwidth of the element. Moreover, the proposed design only needs one single p-i-n diode in each element to realize a 1 bit resolution within a wide operating bandwidth, which can significantly reduce the cost and the complexity of the bias circuit. Besides, the measured peak aperture efficiency is also competitive with other reported designs, due to the low insertion loss of the element.

IV. CONCLUSION

A wideband 1-bit RRA with 2-D dynamic beam-scanning capability has been designed and verified. Based on the ME dipole antenna, a novel wideband RRA element has been developed to achieve wideband operating bandwidth by combining electric dipole resonance and magnetic dipole resonance. A p-i-n diode is integrated into the RRA element to change the types of the resonances, resulting in a 180° reflecting phase difference. To show the advantages of the proposed RRA elements, a 16 × 16 RRA is designed and fabricated by the PCB technology and SMT. In the array design, each RRA element is integrated with a single p-i-n diode to control the reflecting phase individually. Measured results show that the proposed RRA can achieve a peak aperture efficiency of 24% and a 3 dB gain bandwidth of 38.4%, within which wide-angle beam-scanning capability can be obtained. Since the proposed RRA can achieve a wide operating bandwidth, comparative high aperture efficiency, and wide-angle beam-scanning capability, it is a promising candidate to construct a low-cost, large-scale RRA for intelligent reflective surface (IRS) applications in the next-generation wireless communication system.

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